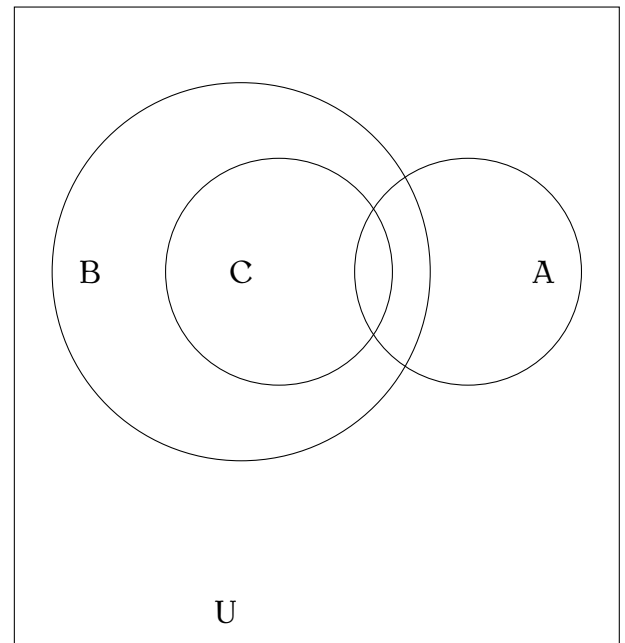
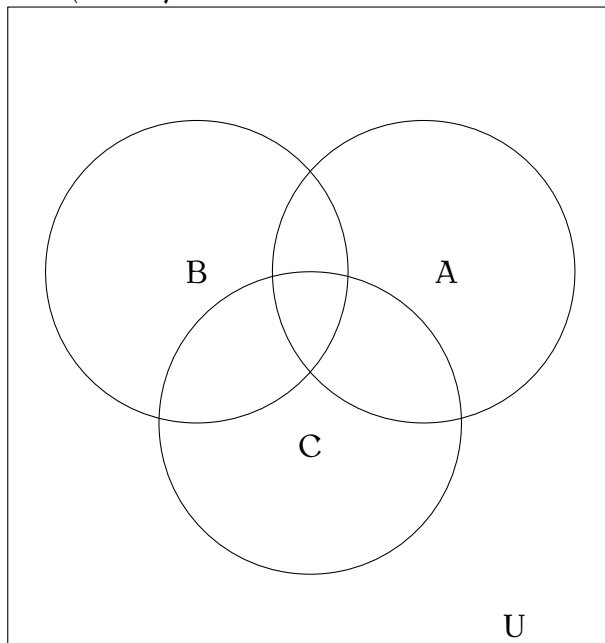
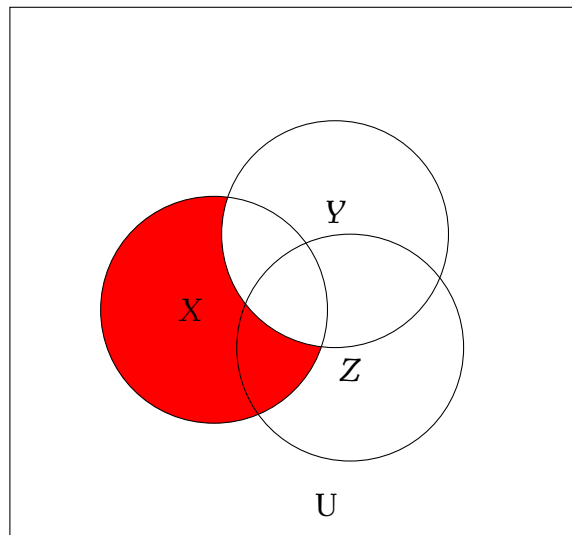
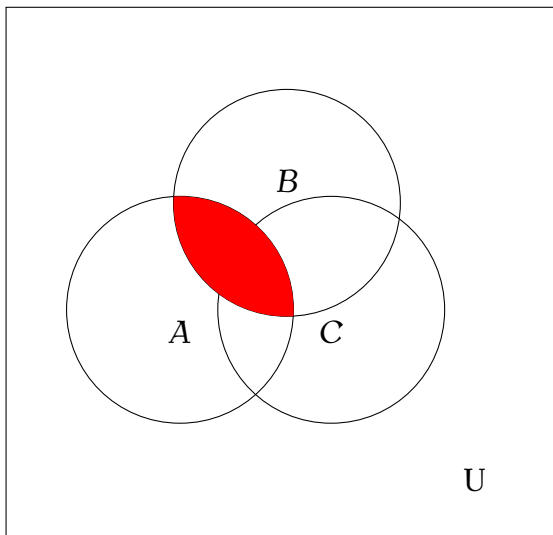


THE UNIVERSITY OF ZAMBIA
 Department of Mathematics and Statistics
 MAT1110: Foundation Mathematics and Statistics for Social Sciences
 Tutorial Sheet 1

1. (a) List the elements of the following sets
- $B = \{x : x \text{ is even}, x < 19, x \in \mathbb{N}\}.$
 - $A = \{x : -2 \leq x < 3, x \in \mathbb{Z}\}$
 - $C = \{x : 4 + x = 3, x \in \mathbb{Z}\}$
 - $D = \{x : x \text{ is a multiple of } 3, x < 10\}.$
 - $E = \{x : x^2 = 1, x \in \mathbb{N}\}.$
 - $F = \{x : x^2 = 4, x \in \mathbb{Z}\}.$
- (b) Let the universal set $U = \{x : x < 10, x \in \mathbb{N}\}$, $A = \{1, 2, 3, 4, 5\}$, $B = \{4, 5, 6, 7\}$, $C = \{5, 6, 7, 8, 9\}$, $D = \{x : x = 2k + 1\}$, $E = \{x : x = 2k, k \in \mathbb{N}\}$, and $F = \{1, 5, 9\}$.
- List down the sets U , D and E , given that D and E are subsets of U . Hence, Find
 - $A \cap (C \cup E).$
 - $B - A$
 - $(A \cap B) - B$
 - $F - (A \cap D)^c$
- (c) Let $U = \{x : x \in \mathbb{N}, 1 \leq x \leq 15\}$. Given that $A = \{x : x = 2k + 1, x < 20, k \in \mathbb{N}\}$ and $B = \{1, 3, 8, 19, 20\}$ and $C = \{1, 2, 6, 8, 9, 11, 15\}$. Illustrate the information on a Venn diagram. Hence, find the following:
- $A^c \cap B^c$
 - $(A \cup B) \cap C$
 - $A - (B \cup C)$
 - $(A \cup B)^c$
- (d) List down all the subsets of each of the following:
- $A = \{a\}$
 - $B = \{a, b\}$
 - $C = \{a, b, c\}$
2. (a) In each of the following Venn diagrams, shade
- $B - C.$
 - $A \cap (B \cap C^c)^c.$
 - $C \cap (A - B)^c.$



- (b) In each of the following Venn diagrams, use the set operations, $\cup$ and $\cap$ to describe the shaded region/s.



3. (a) If $A \subset B$, then simplify each of the following:
- $A \cap B^c$
 - $(B \cup A^c) \cap A$.
- (b) If A and B are disjoint, simplify each of the following:
- $A \cap B$.
 - $A \cup B$.
 - $(A \cap B)^c$
 - $A^c \cup B^c$.
- (c) Express each of the following in its simplest form:
- $(A \cup B) \cap (A \cup B^c)$
 - $[(X \cap Y)^c \cup (X - Y)]^c$.
 - $B^c \cap (A \cup B)$.
 - $[A^c \cup (A \cap B^c)]^c$.
4. (a) Let $\mathbb{R}$ be the universal set. Rewrite each of the following in set builder notation
- $A = [-5, 3)$
 - $B = (4, 7)$
 - $C = [0, 4]$
 - $D = (-2, 6]$
- (b) Let $\mathbb{R}$ be the universal set and $A = (-7, 3]$, $B = (0, 8)$, $C = \{x : x \leq 10, x \in \mathbb{R}\}$. Find each of the following sets and hence illustrate the answers on the number line.
- $A \cap C^c$.
 - $B - A$.
 - $A \cap (B \cup C^c)$.
 - $[A^c - (B - C)]^c$
 - $A^c \cap B^c$.
- (c) Let $A = (-7, 3]$, $B = (0, 8)$. Illustrate De Morgan's laws using sets A and B .